UNIVERSIDAD NACIONAL DE CÓRDOBA
FACULTAD DE CIENCIAS EXACTAS, FÍSICAS Y NATURALES

TEORÍA DE SEÑALES Y SISTEMAS LINEALES

PRÁCTICO: 5

ALEJANDRA LUCÍA PÉREZ LUCERO
GUSTAVO LAZARTE

TRANSFORMADA DE FOURIER PARA SEÑALES DISCRETAS

BIBLIOGRAFÍA

* Señales y Sistemas, Segunda Edición A. V. Oppenheim, A. S. Willsky, S. H. Nawab. Prentice Hall

$$x[n] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X\left(e^{j\omega}\right) e^{j\omega n} \, d\omega$$

TRANSFORMADA DE FOURIER

TRANSFORMADA INVERSA DE FOURIER

$$X\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} x[n] \, e^{-j\omega n}$$

Se puede calcular por integral, tablas y/o propiedades

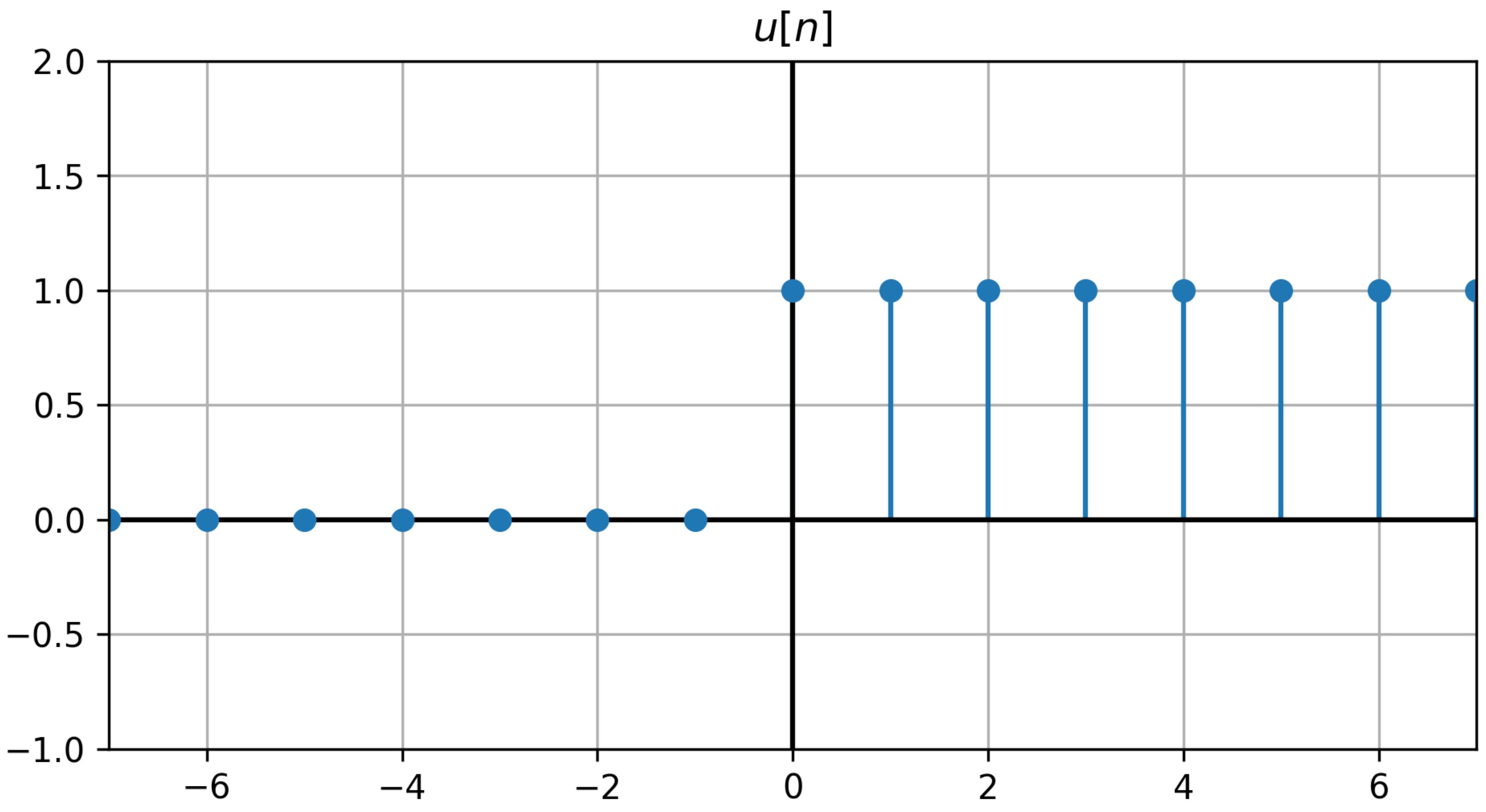
5.21. Calcule la transformada de Fourier de las siguientes señales:

a) $x[n] = u[n-2] - u[n-6]$

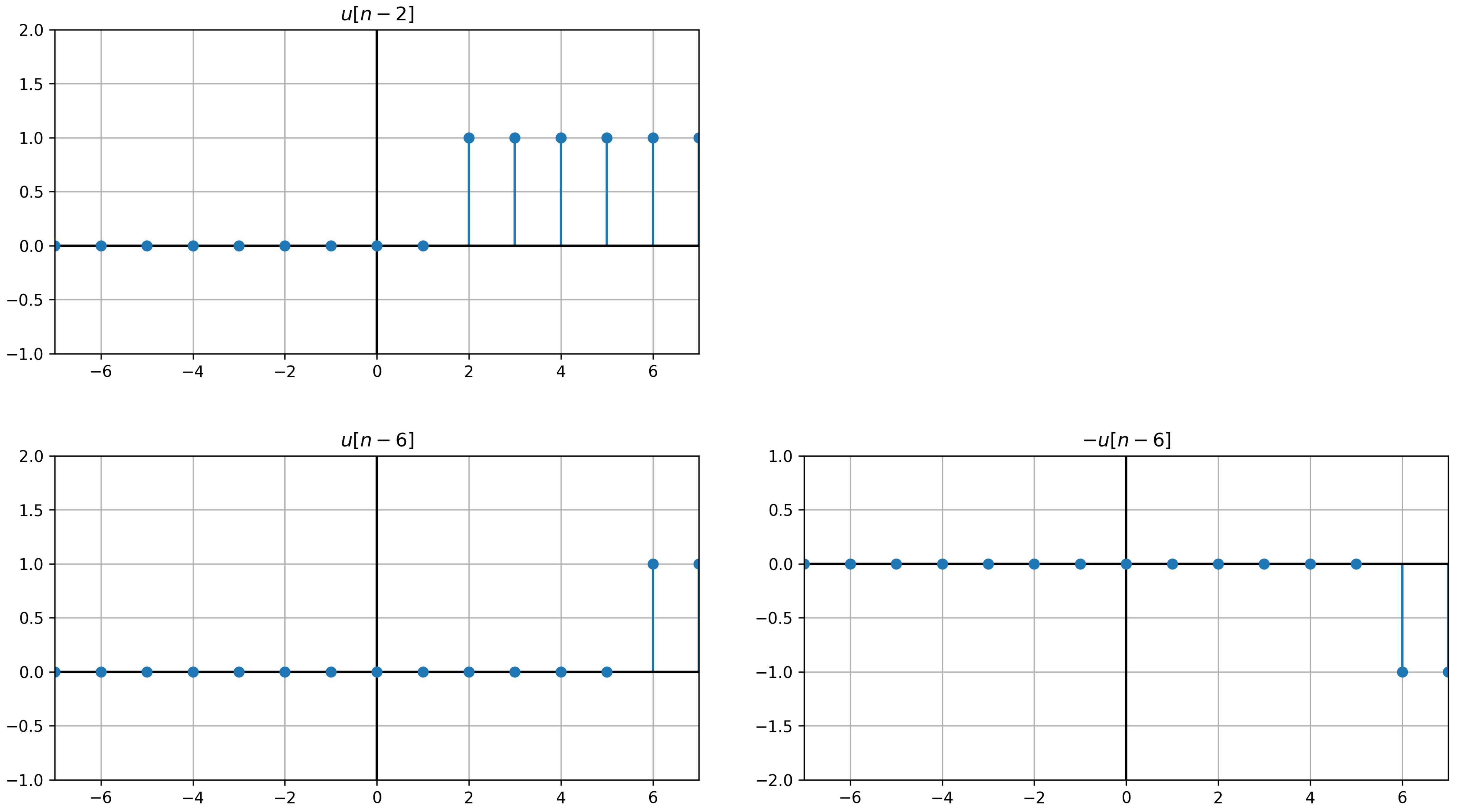
b) $x[n] = \left(\frac{1}{2}\right)^{-n} u[-n-1]$

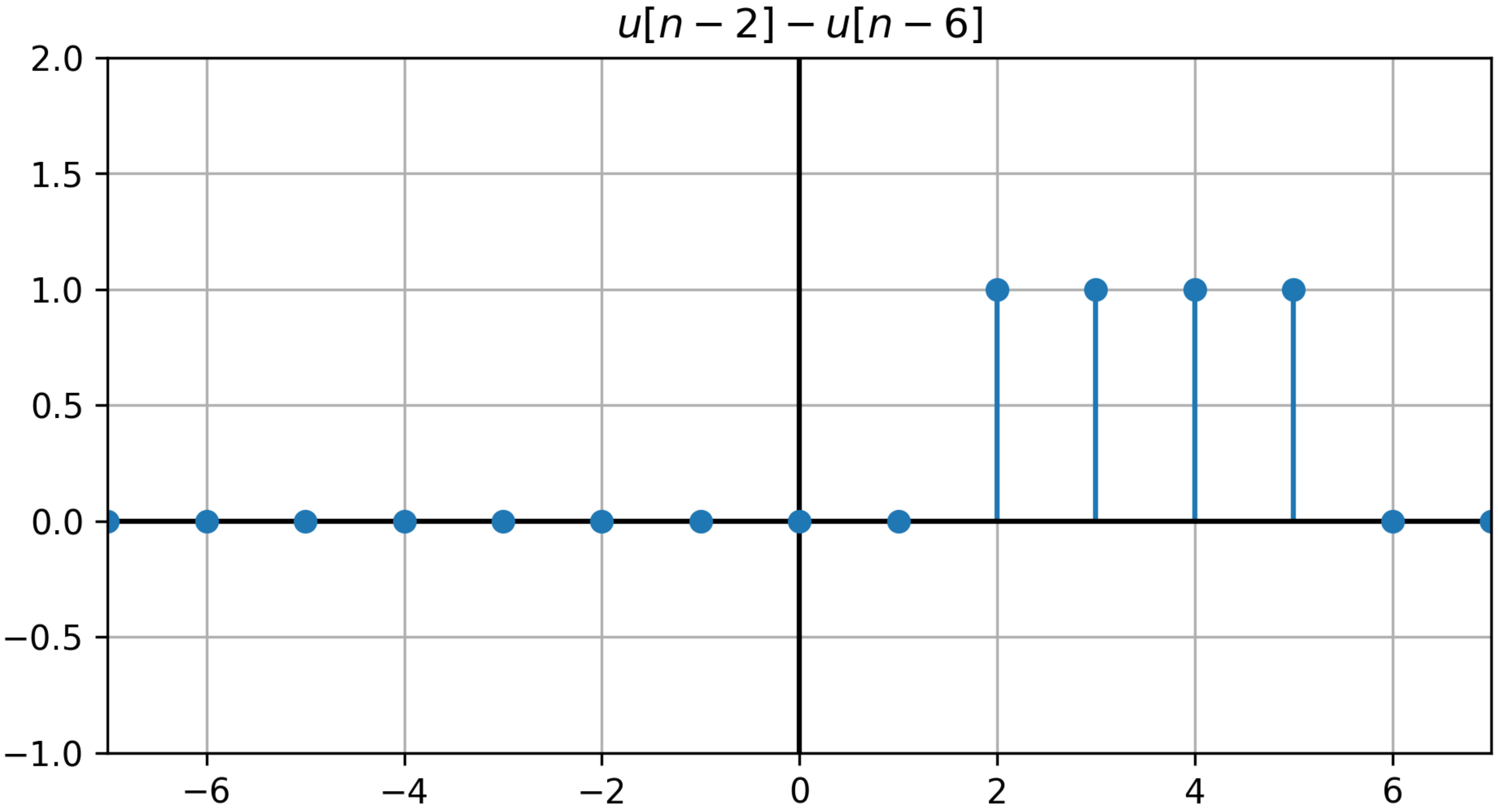
a) $x[n] = u[n-2] - u[n-6]$

Calculo Auxiliar



$$u[n] = \begin{cases} 1 & , n \geq 0 \\ 0 & , n < 0 \end{cases}$$


$u[n-2]$
$u[n-6]$
$-u[n-6]$

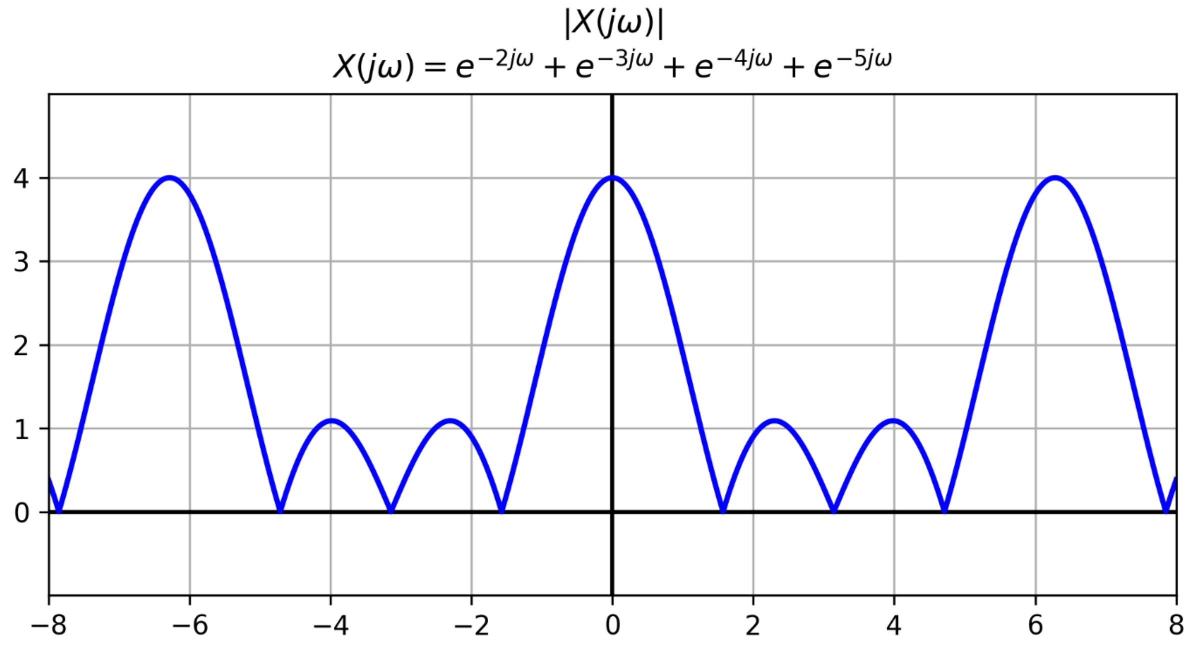

$u[n-2]-u[n-6]$
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−6
−4
−2
0
2
4
6

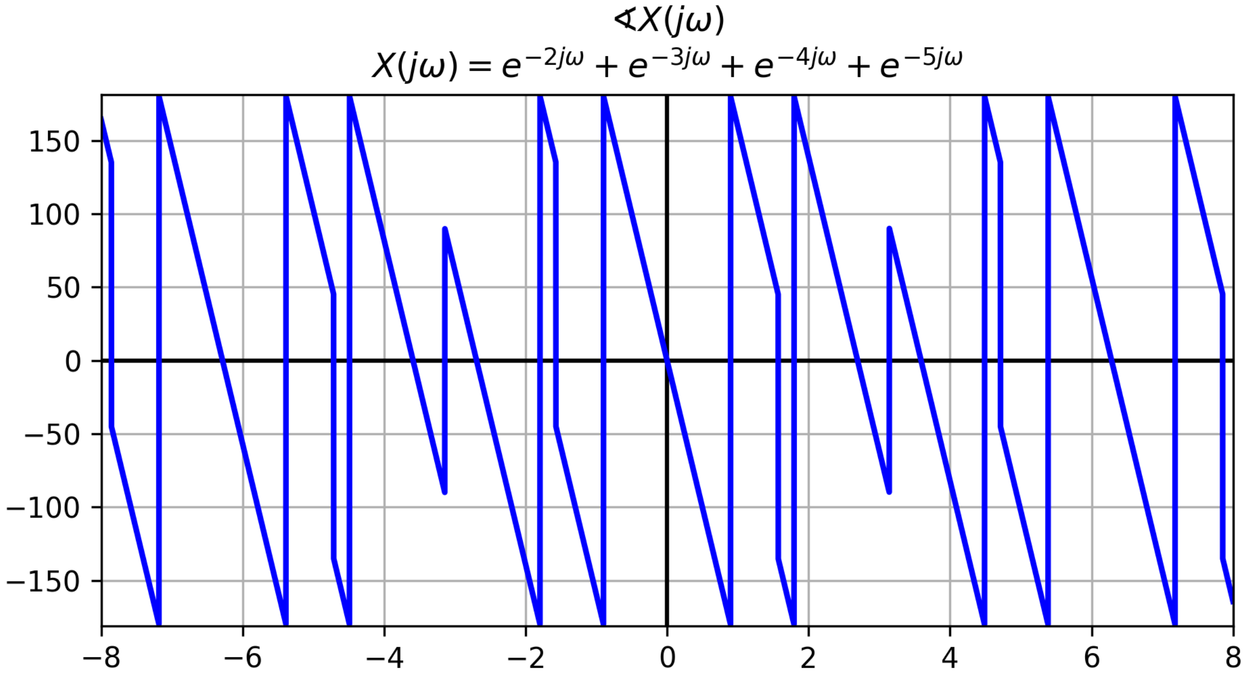
$$x[n] = u[n-2] - u[n-6] = \delta[n-2] + \delta[n-3] + \delta[n-4] + \delta[n-5]$$

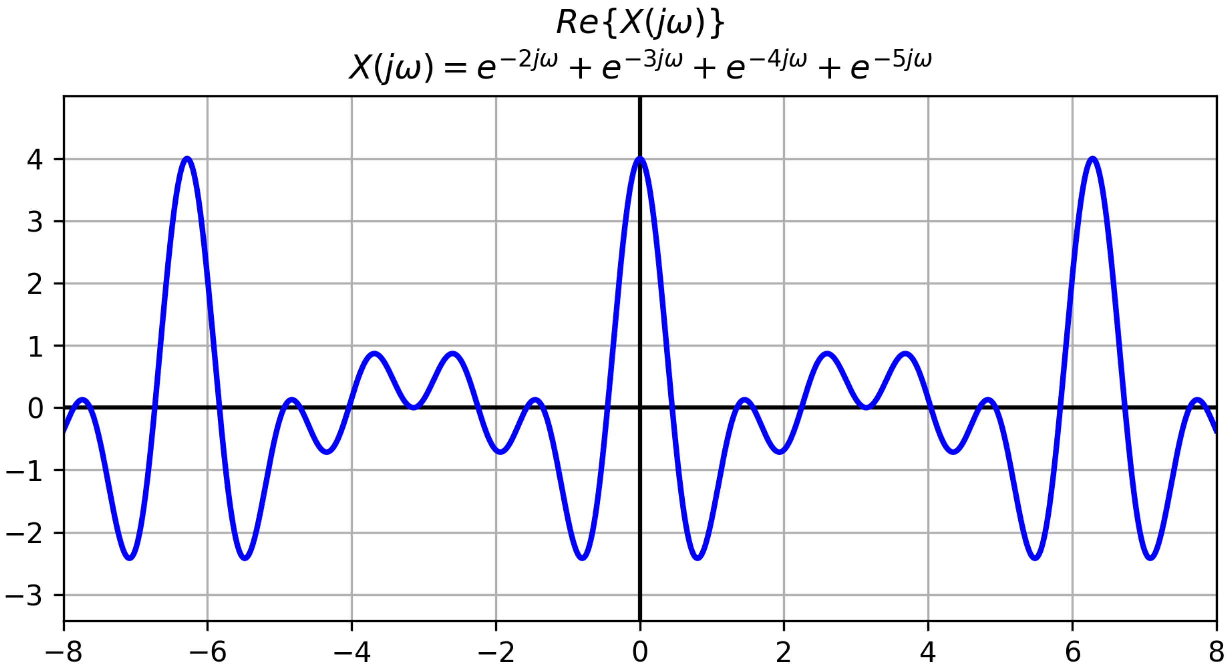
Por tabla:

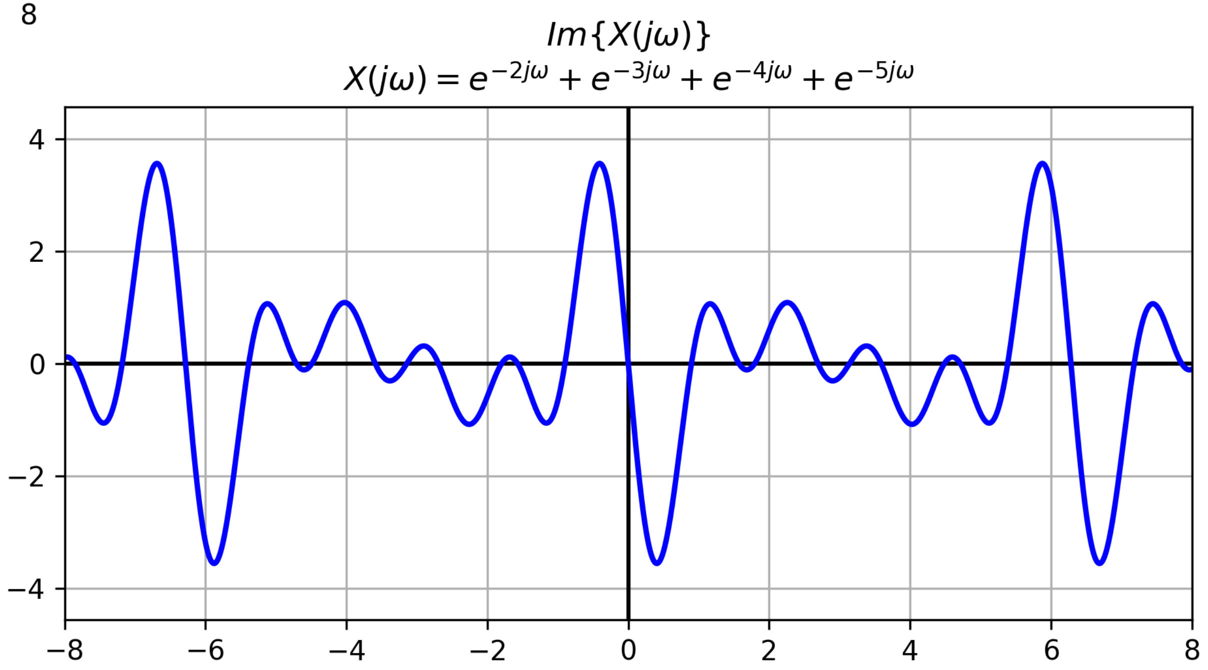
$$\delta[n-n_0] \quad \longleftrightarrow \quad e^{-j\omega n_0}$$

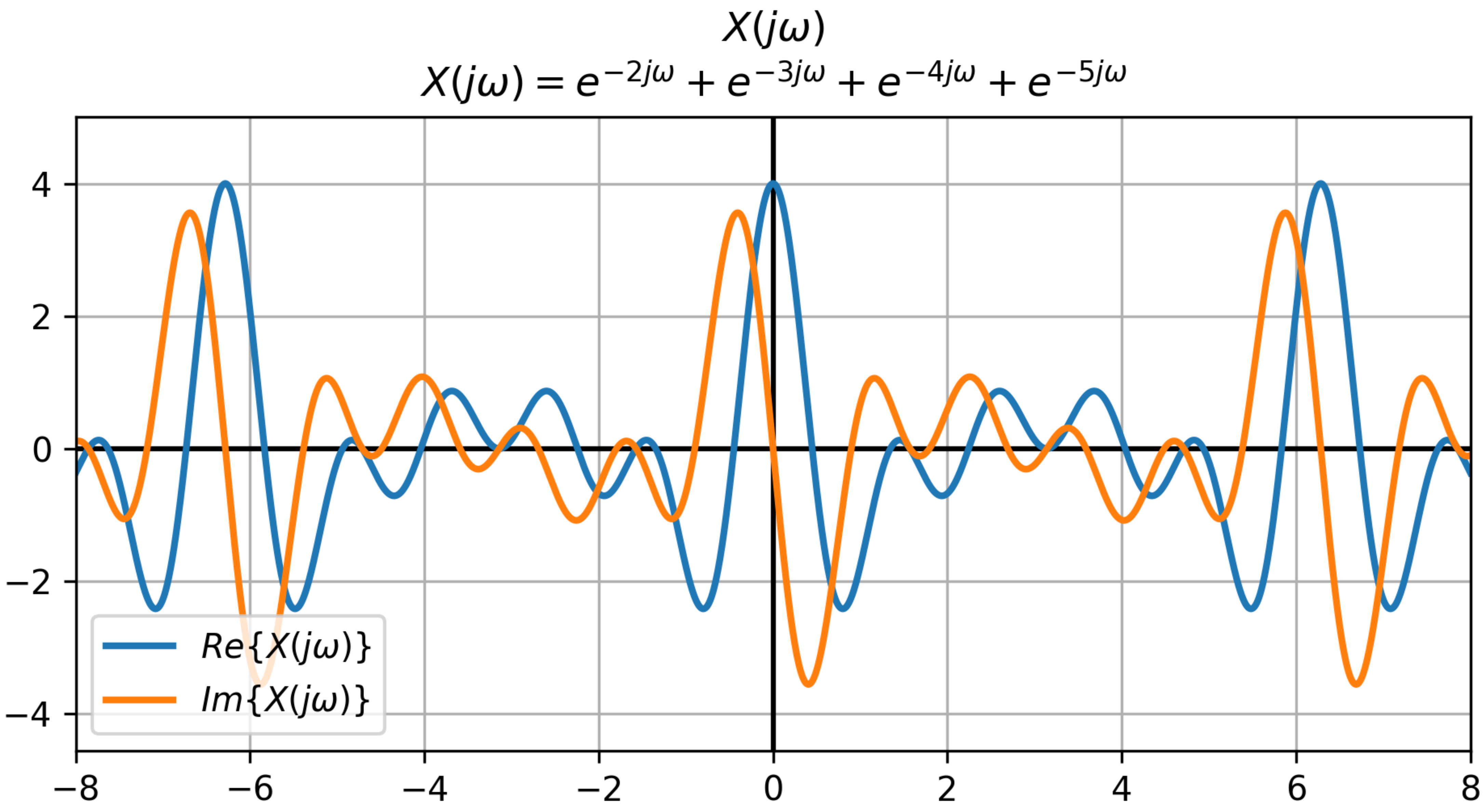
$$X(e^{j\omega}) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$$


$|X(j\omega)|$
$X(j\omega) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$
4
3
2
1
0
−8
−6
−4
−2
0
2
4
6
8


$\measuredangle X(j\omega)$
$X(j\omega) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$
150
100
50
0
−50
−100
−150
−8
−6
−4
−2
0
2
4
6
8


$Re\{X(j\omega)\}$
$X(j\omega) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$
4
3
2
1
0
−1
−2
−3
−8
−6
−4
−2
0
2
4
6
8


$Im\{X(j\omega)\}$
$X(j\omega) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$
4
2
0
−2
−4
−8
−6
−4
−2
0
2
4
6
8

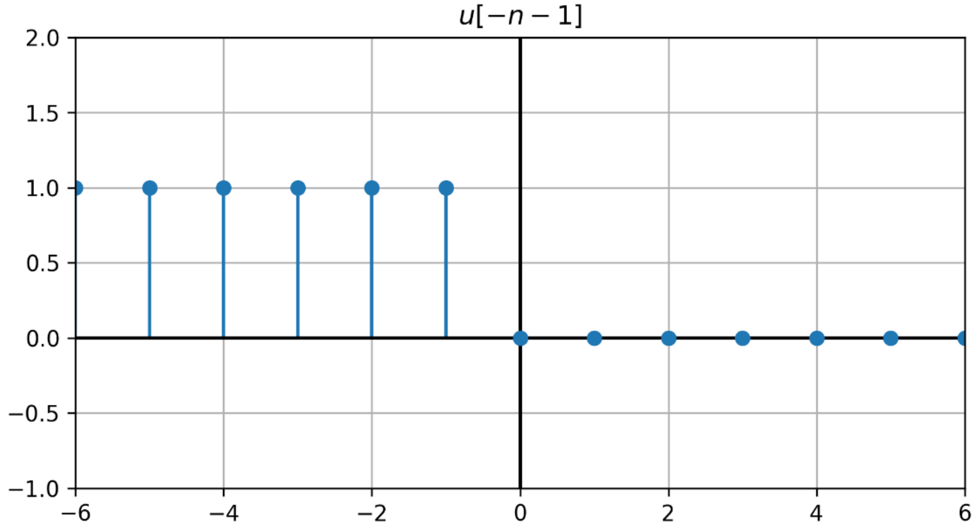


$X(j\omega)$
$X(j\omega) = e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega} + e^{-5j\omega}$
4
2
0
−2
−4
−8
−6
−4
−2
0
2
4
6
8
$Re\{X(j\omega)\}$
$Im\{X(j\omega)\}$

b) $x[n] = \left(\frac{1}{2}\right)^{-n} u[-n-1]$

Solución: $$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$u[-n-1] = u[-(n+1)]$$



$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} \left(\frac{1}{2}\right)^{-n} e^{-j\omega n}$$

Cambiando el signo de "n"

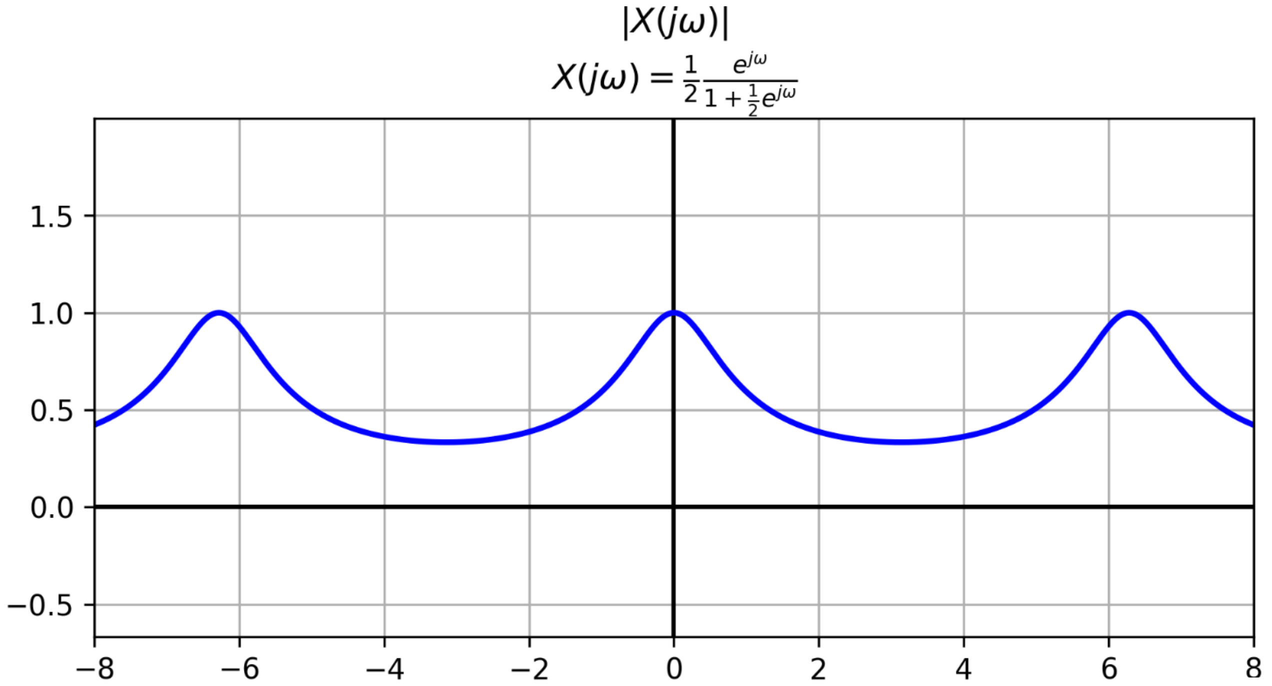
$$X(e^{j\omega}) = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n} e^{j\omega n} = \sum_{n=1}^{\infty} \left(\frac{1}{2} e^{j\omega}\right)^{n}$$

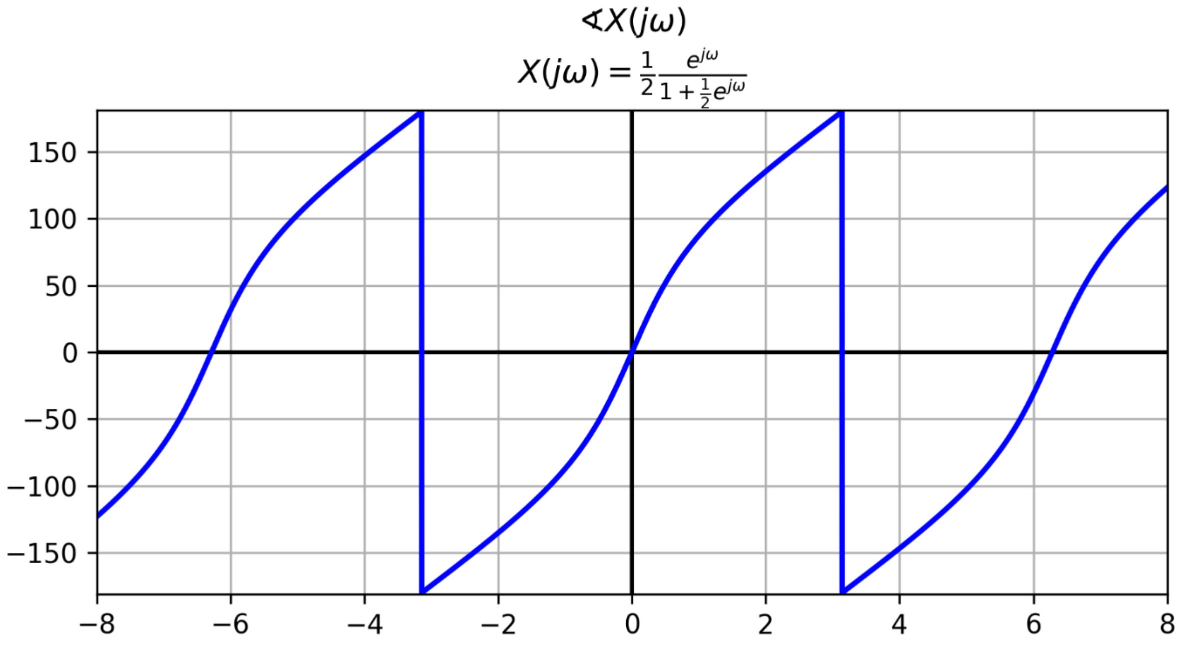
$$X\left(e^{j\omega}\right) = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n} e^{j\omega n} = \sum_{n=1}^{\infty} \left(\frac{1}{2} e^{j\omega}\right)^{n}$$

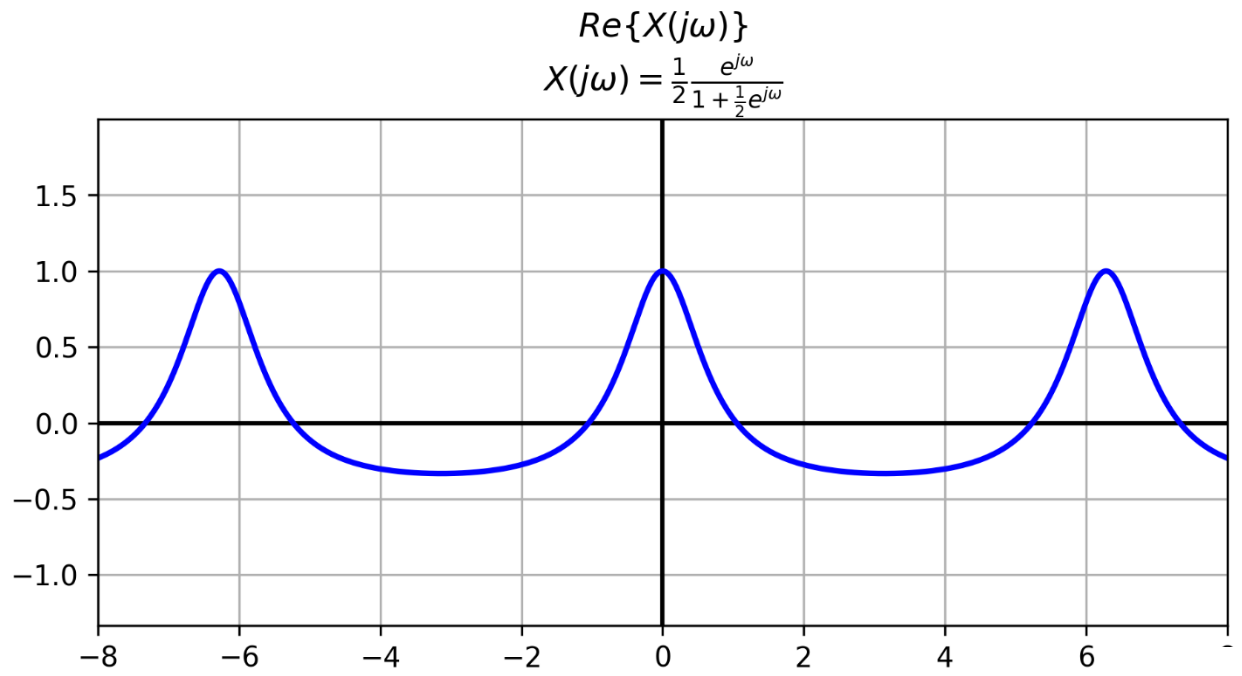
Serie geométrica:

$$\sum_{n=1}^{\infty} \alpha^{n} = \frac{\alpha}{1-\alpha}$$

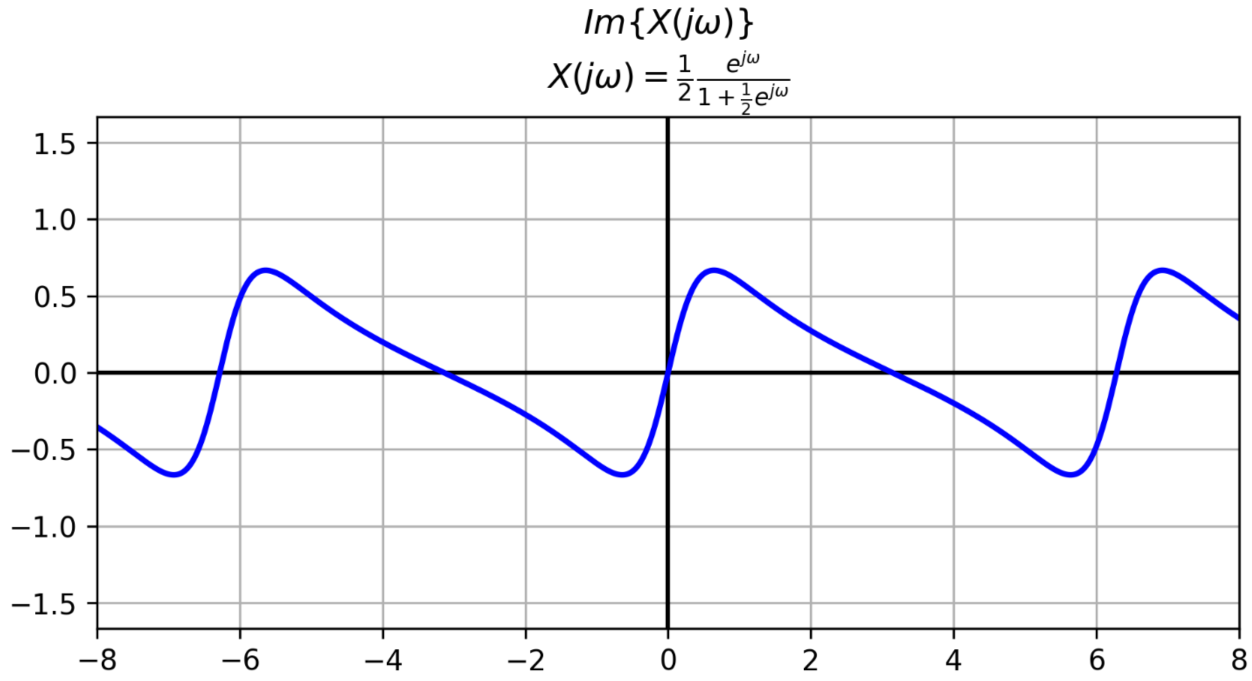
$$X\left(e^{j\omega}\right) = \frac{\frac{1}{2} e^{j\omega}}{1 - \frac{1}{2} e^{j\omega}} = \frac{1}{2} \frac{e^{j\omega}}{\left(1 - \frac{1}{2} e^{j\omega}\right)}$$


$|X(j\omega)|$
$X(j\omega) = \frac{1}{2}\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}}$
1.5
1.0
0.5
0.0
−0.5
−8
−6
−4
−2
0
2
4
6
8

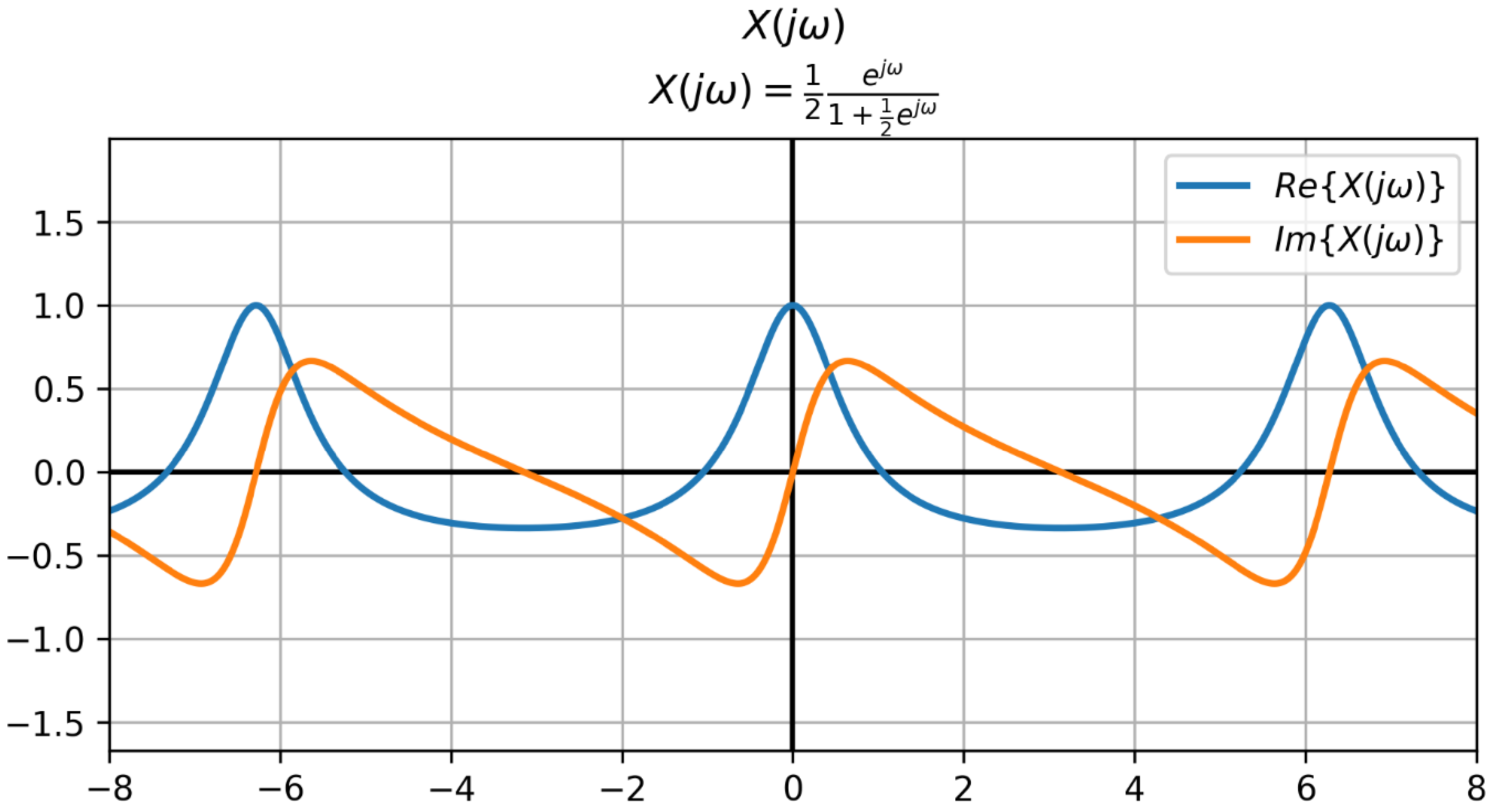

$\measuredangle X(j\omega)$
$X(j\omega) = \frac{1}{2}\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}}$
150
100
50
0
−50
−100
−150
−8
−6
−4
−2
0
2
4
6
8



$Re\{X(j\omega)\}$
$X(j\omega) = \frac{1}{2}\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}}$
1.5
1.0
0.5
0.0
−0.5
−1.0
−8
−6
−4
−2
0
2
4
6



$Im\{X(j\omega)\}$
$X(j\omega) = \frac{1}{2}\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}}$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−8
−6
−4
−2
0
2
4
6
8


$X(j\omega)$
$X(j\omega) = \frac{1}{2}\frac{e^{j\omega}}{1+\frac{1}{2}e^{j\omega}}$
$Re\{X(j\omega)\}$
$Im\{X(j\omega)\}$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−8
−6
−4
−2
0
2
4
6
8

5.22. A continuación se muestran las transformadas de Fourier de las señales discretas. Determine la señal correspondiente a cada transformada.

$$X(e^{j\omega}) = \begin{cases} 1, & \dfrac{\pi}{4} \leq |\omega| \leq \dfrac{3\pi}{4} \\ 0, & \dfrac{3\pi}{4} \leq |\omega| \leq \pi \,, \quad 0 \leq |\omega| \leq \dfrac{\pi}{4} \end{cases}$$

Antitransformada de Fourier:

$$x[n] = \frac{1}{2\pi} \int_{-\infty}^{\infty} X\left(e^{j\omega}\right) e^{j\omega n} \, d\omega$$

Calculo Auxiliar

$$\int e^{ax} \, dx = \frac{1}{a} \, e^{ax}$$

$$x[n] = \frac{1}{2\pi} \int_{-\frac{3\pi}{4}}^{-\frac{\pi}{4}} e^{j\omega n} \, d\omega + \frac{1}{2\pi} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} e^{j\omega n} \, d\omega$$

$$x[n] = \frac{1}{2\pi} \left[\left| \frac{e^{j\omega n}}{jn} \right|_{-\frac{3\pi}{4}}^{-\frac{\pi}{4}} + \left| \frac{e^{j\omega n}}{jn} \right|_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \right]$$

$$x[n] = \frac{1}{2\pi}\left[\left(\frac{e^{-j\frac{\pi}{4}n} - e^{-j\frac{3\pi}{4}n}}{jn}\right) + \left(\frac{e^{j\frac{3\pi}{4}n} - e^{j\frac{\pi}{4}n}}{jn}\right)\right]$$

$$x[n] = \frac{1}{2\pi n}\left[2\left(\frac{e^{-j\frac{\pi}{4}n} - e^{-j\frac{3\pi}{4}n}}{2j}\right) + 2\left(\frac{e^{j\frac{3\pi}{4}n} - e^{j\frac{\pi}{4}n}}{2j}\right)\right]$$

$$x[n] = \frac{1}{\pi n}\left[\sin\left(\frac{3\pi}{4}n\right) - \sin\left(\frac{\pi}{4}n\right)\right]$$

5.23. Sea que $X\left(e^{j\omega}\right)$ que denota la transformada de Fourier de la señal $x[n]$ representada en la figura. Realice los siguientes cálculos sin evaluar explícitamente $X\left(e^{j\omega}\right)$:

a) Evalúe $X\left(e^{jo}\right)$.

b) Encuentre $\sphericalangle X\left(e^{j\omega}\right)$.

c) Evalúe $\int_{-\pi}^{\pi} X\left(e^{j\omega}\right) d\omega$.

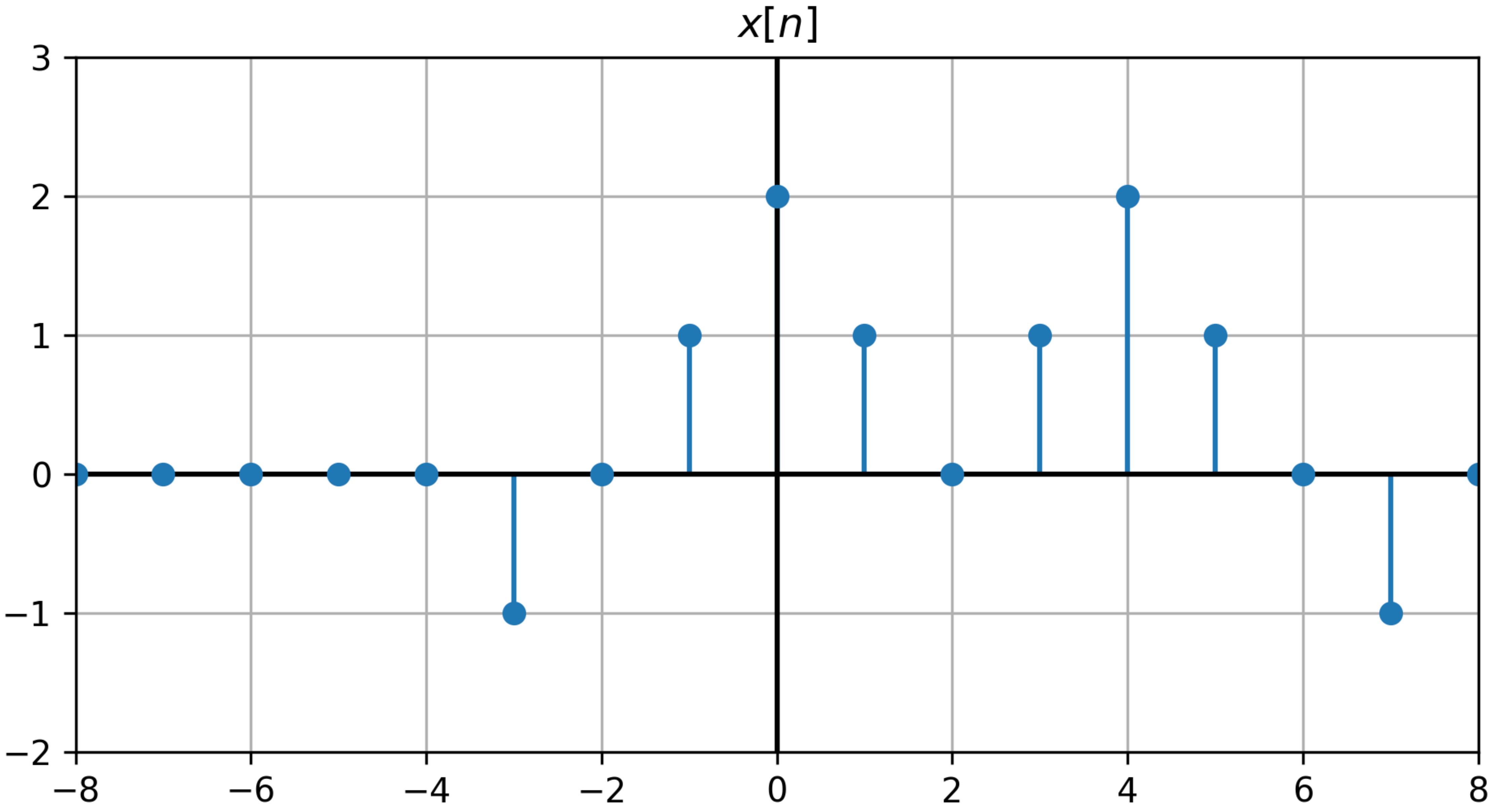
d) Encuentre $X\left(e^{j\pi}\right)$.

e) Determine y dibuje la señal cuya transformada de Fourier es $\mathcal{R}e\{x(\omega)\}$

f) Evalúe

i) $\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$

ii) $\int_{-\pi}^{\pi} \left|\frac{dX\left(e^{j\omega}\right)}{d\omega}\right|^2 d\omega$


x[n]
3
2
1
0
−1
−2
−8
−6
−4
−2
0
2
4
6
8

a) Evalúe $X(e^{jo})$.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$X(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n] = \sum_{n=-3}^{7} x[n]$$

$$X(e^{j0}) = \quad x[-3] + x[-2] + x[-1] + x[0] + x[1] + x[2] + x[3] + x[4] +$$

$$+x[5] + x[6] + x[7]$$

$$X(e^{j0}) = -1 + 1 + 2 + 1 + 1 + 2 + 1 - 1 = 6$$

b) Encuentre $\measuredangle X(e^{j\omega})$.

Vemos que : $y[n] = x[n+2]$ es una señal par.

Entonces la transformada de Fourier: $Y(e^{j\omega})$ es real y par $\Rightarrow \measuredangle Y(e^{j\omega}) = 0$

Por propiedad de "desplazamiento en el tiempo"

$$Y(e^{j\omega}) = e^{j2\omega} X(e^{j\omega})$$

Entonces:

$$\measuredangle Y(e^{j\omega}) = \measuredangle e^{j2\omega} + \measuredangle X(e^{j\omega})$$

$$0 = \measuredangle e^{j2\omega} + \measuredangle X(e^{j\omega})$$

$$\therefore \quad \measuredangle X(e^{j\omega}) = -\measuredangle e^{j2\omega} = \measuredangle e^{-j2\omega}$$

Calculo Auxiliar

$$e^{j2\omega} = \cos 2\omega + j \sin 2\omega$$

$$e^{j2\omega} = r \measuredangle\theta = r \angle\theta$$

$$r = [cos^2(2\omega) + sin^2(2\omega)]^{\frac{1}{2}} = 1$$

$$\theta = tg^{-1}\frac{\sin 2\omega}{\cos 2\omega} = tg^{-1} tg\,(2\omega)$$

$$\theta = 2\omega$$

c) Evalúe $\int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega$.

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$2\pi\, x[n] = \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$2\pi\, x[0] = \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega 0}\, d\omega$$

$$2\pi\, 2 = \int_{-\pi}^{\pi} X(e^{j\omega})\, d\omega = 4\pi$$

d) Encuentre $X(e^{j\pi})$.

Calculo Auxiliar

$$e^{-j\pi} = \cos(-\pi) + j\sin(-\pi)$$

$$e^{-j\pi} = -1$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\pi n} = \sum_{n=-\infty}^{\infty} x[n]\, \left(e^{-j\pi}\right)^{n}$$

$$X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} x[n]\, (-1)^{n} = \sum_{n=-3}^{7} x[n]\, (-1)^{n}$$

$$X\left(e^{j\pi}\right) = \sum_{n=-3}^{7} x[n]\,(-1)^n$$

$$X\left(e^{j\pi}\right) = x[-3](-1)^{-3} + x[-1](-1)^{-1} + x[0](-1)^{0} + x[1](-1)^{1} +$$

$$+x[3](-1)^{3} + x[4](-1)^{4} + x[5](-1)^{5} + x[7](-1)^{7}$$

$$X\left(e^{j\pi}\right) = \ (-1)(-1) + 1\,(-1) + 2\,.\,1 + 1.\,(-1) + 1\,.\,(-1) + 2.\,1 +$$

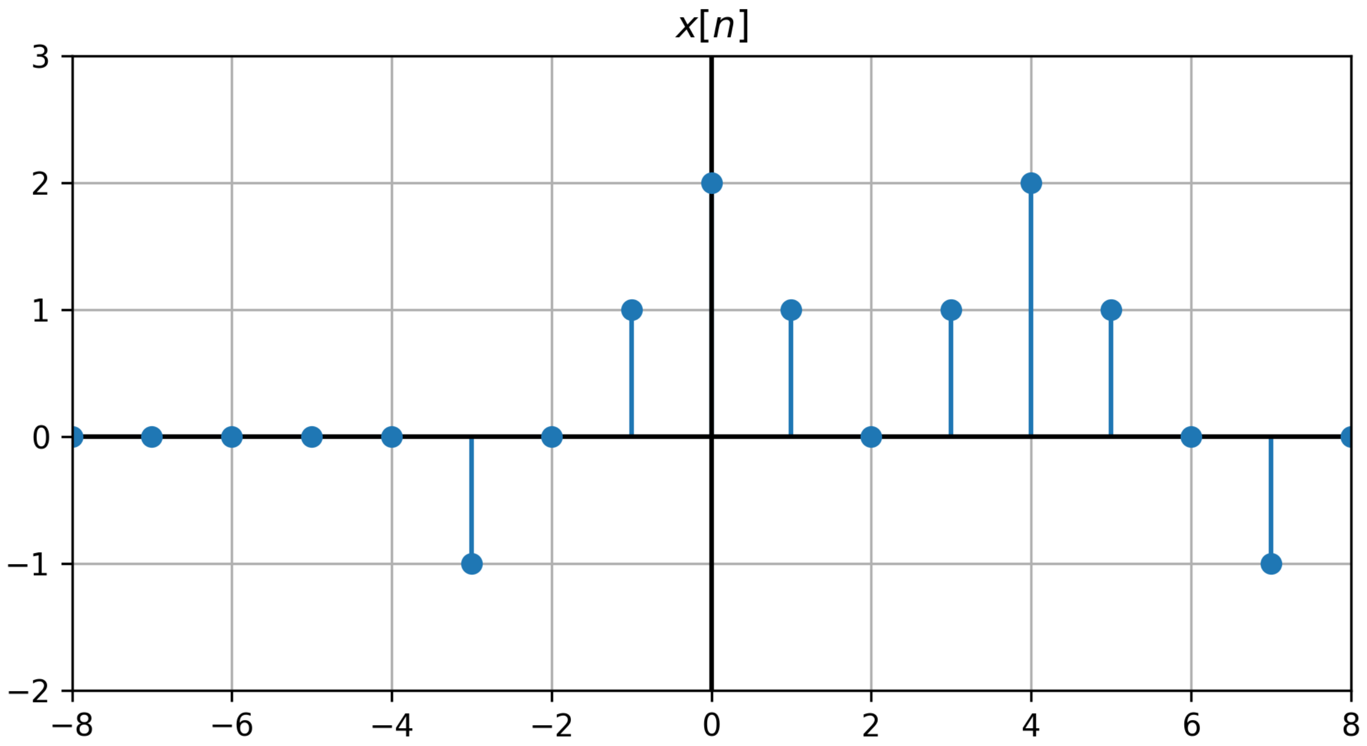
$$+1.\,(-1) + (-1)(-1)$$

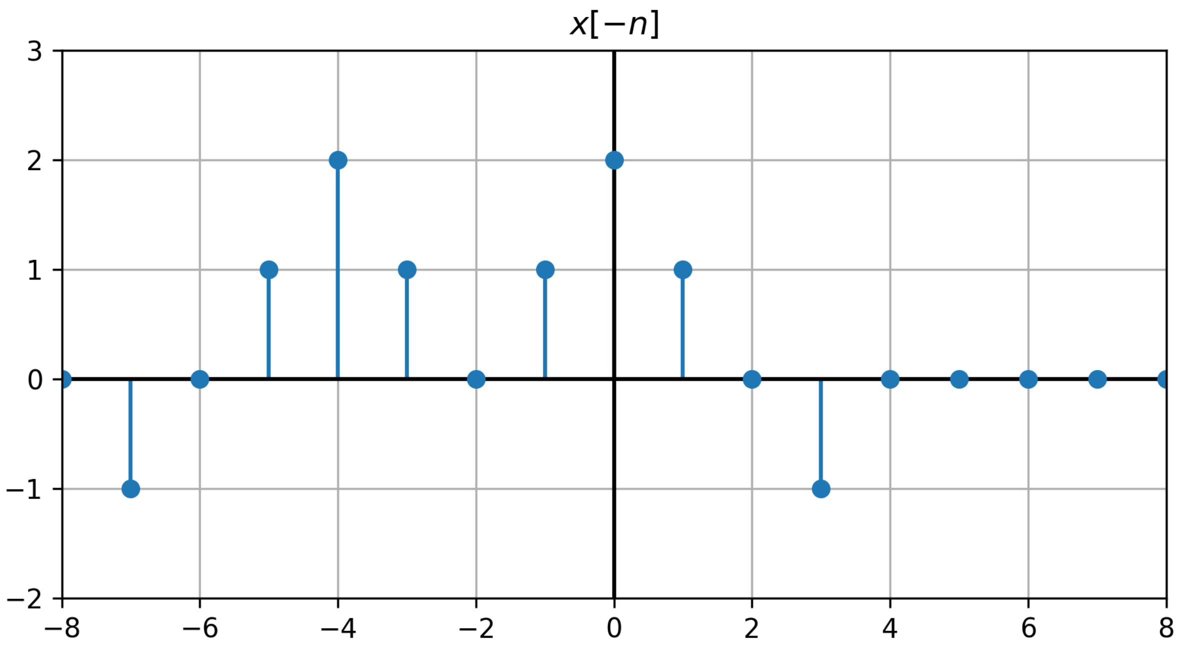
$$X\left(e^{j\pi}\right) = 1 - 1 + 2 - 1 - 1 + 2 - 1 + 1 = 2$$

e) Determine y dibuje la señal cuya transformada de Fourier es $\mathcal{Re}\{X(\omega)\}$

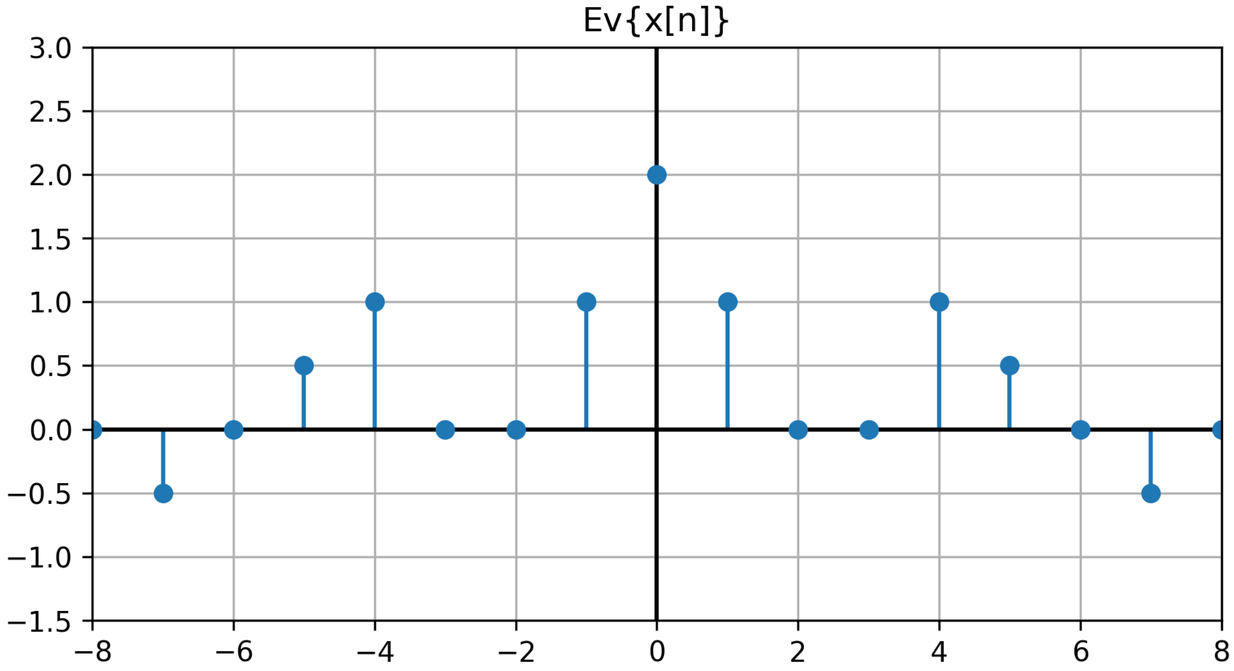
Por tabla (descomposición de señales reales en par e impar)

$$X_e[n] = \varepsilon v\,\{X[n]\} \quad \longleftrightarrow \quad \mathcal{Re}\{X(e^{j\omega})\} \qquad y \qquad \varepsilon v\,\{X[n]\} = \frac{x[n]+x[-n]}{2}$$





$x[-n]$
3
2
1
0
−1
−2
−8
−6
−4
−2
0
2
4
6
8



Ev{x[n]}
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−8
−6
−4
−2
0
2
4
6
8

f) Evalúe

i) $\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$

Relación de Parseval para señales aperiódicas

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$$

$$2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2 = \int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$$

$$2\pi \sum_{n=-3}^{7} |x[n]|^2 = \int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$$

$$\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega = 2\pi \sum_{n=-3}^{7} |x[n]|^2$$

$$\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega = 2\pi(x^2[-3] + x^2[-2] + x^2[-1] + x^2[0] + x^2[1] + \\ + x^2[2] + x^2[3] + x^2[4] + x^2[5] + x^2[6] + x^2[7])$$

$$\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega = 2\pi[1 + 1 + 4 + 1 + 1 + 4 + 1 + 1]$$

$$\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega = 28\,\pi$$

ii) $\int_{-\pi}^{\pi}\left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 \, d\omega$

Por propiedad “diferenciación en frecuencia”

$$n\,x[n] \qquad \longleftrightarrow \qquad j\frac{dX(e^{j\omega})}{d\omega}$$

Usando el Teorema de Parseval:

$$\int_{-\pi}^{\pi}\left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 \, d\omega = 2\pi \sum_{n=-\infty}^{\infty} |n|^2 |x[n]|^2$$

$$\int_{-\pi}^{\pi}\left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 \, d\omega = 2\pi \sum_{n=-3}^{7} |n|^2 |x[n]|^2$$

$$\int_{-\pi}^{\pi} \left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 d\omega = 2\pi \sum_{n=-3}^{7} |n|^2 |x[n]|^2$$

$$\int_{-\pi}^{\pi} \left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 d\omega = 2\pi(|-3|^2|x[-3]|^2 + |-2|^2|x[-2]|^2 + |-1|^2|x[-1]|^2 + \\ +|0|^2|x[0]|^2 + |1|^2|x[1]|^2 + |2|^2|x[2]|^2 + |3|^2|x[3]|^2 + \\ +|4|^2|x[4]|^2 + |5|^2|x[5]|^2 + |6|^2|x[6]|^2 + |7|^2|x[7]|^2)$$

$$\int_{-\pi}^{\pi} \left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 d\omega = 2\pi(9.1 + 4.0 + 1.1 + 0 + 1.1 + 0 + 9.1 + 16.4 + 25.1 + \\ +0 + 49.1)$$

$$\int_{-\pi}^{\pi} \left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 d\omega = 2\pi(9 + 1 + 1 + 9 + 64 + 25 + 49) = 316\,\pi$$